

1. Initial Incident Summary

At **02:14 AM**, the SOC received multiple alerts indicating a possible cyberattack. Several failed login attempts from the IP **185.23.44.91** were followed by a successful login to the **backup_admin** account. The attacker then executed an **encoded PowerShell command**, created a **new local administrator account**, connected to an external IP (**45.77.122.18**), and transferred **3.5 GB of data** outside the network. These events strongly suggest a successful system compromise and possible data theft.

2. Attack Timeline

Event	Description
Failed Logins	Multiple Event ID 4625 failed login attempts from an external IP.
Successful Login	The backup_admin account logged in successfully.
PowerShell Execution	An encoded PowerShell command was executed.
Privilege Escalation	A new local administrator account was created.
External Connection	The system connected to 45.77.122.18 .
Data Exfiltration	Around 3.5 GB of data was transferred outside the network.

3. Possible Attack Chain

The attacker likely performed a **brute-force attack** to gain access to the **backup_admin** account. After logging in, they executed an encoded PowerShell command, created a new administrator account for persistence, connected to a Command and Control (C2) server, and finally exfiltrated sensitive data.

4. MITRE ATT&CK Mapping

Technique	MITRE ID
Brute Force	T1110
Valid Accounts	T1078

PowerShell	T1059.001
Create Account	T1136
Command and Control	T1071
Data Exfiltration	T1041

5. Indicators of Compromise (IOCs)

- Source IP: **185.23.44.91**
- Destination IP: **45.77.122.18**
- Compromised Account: **backup_admin**
- Encoded PowerShell execution
- New administrator account creation
- 3.5 GB outbound data transfer

6. Risk Assessment

Risk Level: Critical

The attacker gained administrator access and transferred a large amount of data outside the network. This could lead to data loss, unauthorized access, and serious business impact if not contained quickly.

7. Containment Actions

- Isolate the affected system from the network.
- Disable the compromised **backup_admin** account.
- Remove the unauthorized administrator account.
- Block the malicious IP addresses.
- Reset passwords and enable MFA.
- Preserve logs and forensic evidence for investigation.

8. Detection Opportunities

- Alert on repeated failed login attempts.
- Monitor encoded PowerShell execution.
- Detect new administrator account creation.
- Monitor large outbound data transfers.

- Use threat intelligence to identify malicious IP addresses.

9. Executive Summary

The investigation indicates a successful cyberattack where the attacker compromised an administrator account, executed malicious PowerShell commands, established persistence, communicated with an external server, and exfiltrated 3.5 GB of data. Immediate containment and further investigation are recommended.

10. Final Verdict

This incident should be classified as a **Critical Security Incident**. The evidence suggests credential compromise, privilege escalation, persistence, and data exfiltration. Immediate incident response and forensic analysis are required to prevent further damage.